Prayas JEE AIR (2025)

WEEKLY TEST 32

DURATION : 180 Minutes | **DATE : 18/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

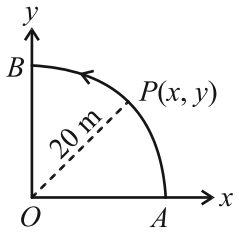
OMR Bar Code Number: _____

Candidate's Signature: _____ Invigilator's Signature _____

SECTION-I (PHYSICS)

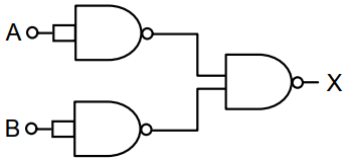
Single Correct Type Questions

1. A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a lengths $s = t^3 + 5$, where s is in metres an t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 2$s is nearly



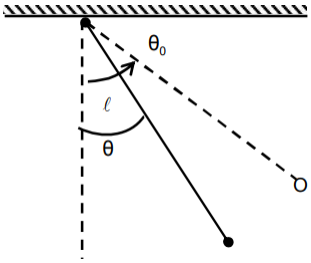
(1) 13 m/s^2 (2) 12 m/s^2
(3) 7.2 m/s^2 (4) 14 m/s^2

2. The combination of gates shown below is



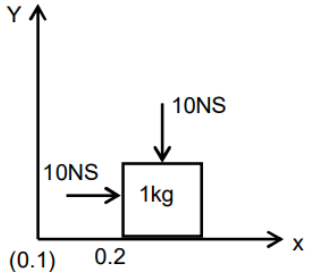
(1) OR gate (2) NOT gate
(3) XOR gate (4) NAND gate

3. A point mass is attached to an inextensible string of length ℓ. The particle is released from a position where the string is initially horizontal. At some time string makes an angle θ with vertical & let u be the speed of particle at that instant then (T is tension in string)



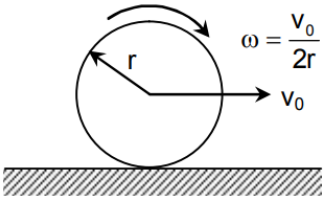
(1) $T\cos\theta = mg$ (2) $mg\cos\theta = T$
(3) $T = 3mg\cos\theta$ (4) $T = 2mg\cos\theta$

4. A block of mass 1 kg is resting on a rough horizontal surface with coefficient of friction 0.2. If an impulse $(10\hat{i} - 10\hat{j})$ N-s is given to block then velocity of block just after the given impulse is



(1) 10 m/sec (2) 0 m/sec
(3) 8 m/sec (4) 4 m/sec

5. A solid sphere of mass M and radius r slips on a rough horizontal surface. At some instant, it has translational velocity v_0 and angular velocity $\omega = \frac{v_0}{2r}$ about its center as shown in figure. The translational velocity after the sphere starts pure rolling motion is

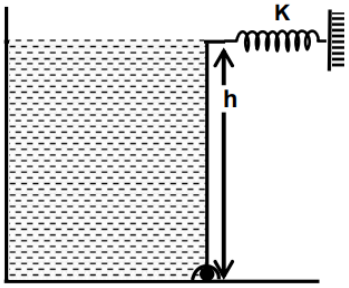


(1) $\frac{v_0}{7}$ (2) $\frac{2v_0}{7}$
(3) $\frac{5v_0}{7}$ (4) $\frac{6v_0}{7}$

6. The acceleration due to gravity on the surface of the moon is $\frac{1}{6}$ that of the surface of earth and the diameter of the moon is one-fourth that of earth. The ratio of escape velocity on earth to escape velocity on moon will be:

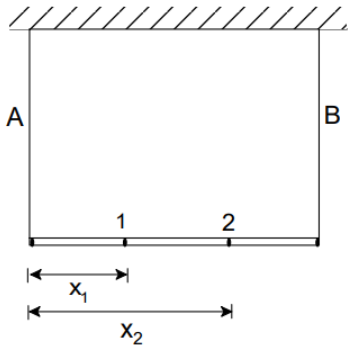
(1) $\frac{\sqrt{6}}{2}$ (2) $\sqrt{24}$
(3) 3 (4) $\frac{\sqrt{3}}{2}$

7. A plank of length h and width b (perpendicular to plane of paper) is hinged at one end. The upper end is attached with a spring of spring constant k and water is filled up to high h. If plank is in equilibrium in vertical position then potential energy stored in spring is (ignore effect of atmospheric pressure, density of water is ρ and acceleration due to gravity is g)



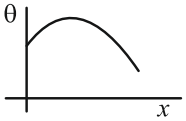
(1) $\frac{b^2\rho^2g^2h^4}{18}$ (2) $\frac{b^2\rho^2g^2h^4}{36k}$
(3) $\frac{b^2\rho^2g^2h^4}{72k}$ (4) $\frac{b^2\rho^2g^2h^4}{128k}$

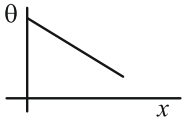
8. A light rigid rod of length 'ℓ' is suspended at its two ends by two wires 'A' and 'B ' of equal lengths and area of cross section 'a' and '$2a$' respectively. ($Y_A = Y_0$ & $Y_B = 3Y_0$)
A mass 'm' is suspended at points '1' and '2', one by one. When it is suspended at point '1', stress developed in both wires is same; and when it is suspended at point '2', strain developed in both wires is same. Ratio of $x_1 : x_2$ is given by

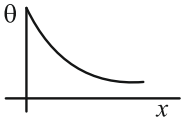


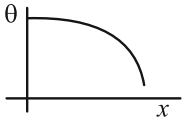
(1) 7 : 18 (2) 7 : 9
(3) 7 : 3 (4) 14 : 3

9. A long metallic bar is having constant rate of heat flow from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figure.

(1) 

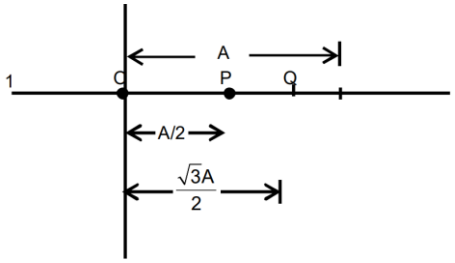
(2) 

(3) 

(4) 

10. One mole of ideal monoatomic gas (γ = 5/3) is mixed with one mole of diatomic gas (γ = 7/5). What is γ for the mixture? γ denotes the ratio of specific heat at constant pressure, to that at constant volume.
(1) 3/2 (2) 23/15
(3) 35/23 (4) 4/3

11. A particle is performing simple harmonic motion with amplitude A and time period T. Time taken by particle in going from point P to Q is

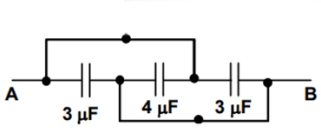


(1) $T/3$ (2) $T/6$
(3) $T/12$ (4) $T/18$

12. If M_O is the mass of an oxygen isotope ${}_8O^{17}$, M_P and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is
(1) $(M_O - 8M_P)C^2$
(2) $(M_O - 8M_P - 9M_N)\,C^2$
(3) $(M_O - 9M_P - 8M_N)\,C^2$
(4) $(M_O - 17M_N)\,C^2$

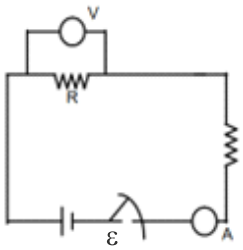
13. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_o\left(\frac{5}{4} - \frac{r}{R}\right)$ upto $r = R$ and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance r ($r < R$) from the origin is given by
(1) $\frac{4\pi\rho_o r}{3\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$
(2) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$
(3) $\frac{4\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4} - \frac{r}{R}\right)$
(4) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4} - \frac{r}{R}\right)$

14. The equivalent capacitance of the circuit between A and B is



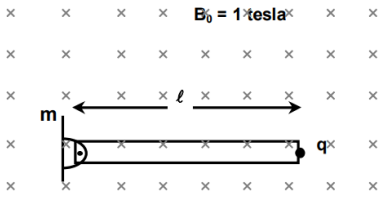
(1) 10 μF (2) 5 μF
(3) 15 μF (4) 20 μF

15. The potential deference across the resistor as measured by an ideal voltmeter is 60 ± 1.5 V and the current through the resistor as measured by an ideal ammeter is (2 ± 0.01) A. what is the maximum % error in the value of R?



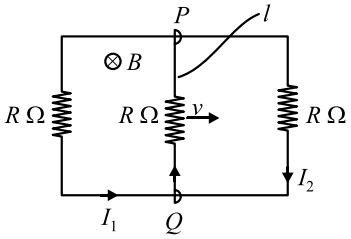
(1) 3%
(2) 1.5%
(3) 2.5%
(4) None of these

16. A uniform non conducting bar of mass 1 kg, length 1 m is hinged at one end so that it can rotate freely in vertical plane. A charge $\sqrt{\frac{3}{10}}$C is placed on the other end. If the bar is released from horizontal position the force exerted on the hinge, when the rod becomes vertical, is



(1) 22 N (2) 38 N
(3) 28 N (4) 18 N

17. A rectangular loop has a sliding connector *PQ* of length ℓ and resistance $R\ \Omega$ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field B going into the plane of the paper. The three currents I_1, I_2 and I are



(1) $I_1 = -I_2 = \frac{B\ell v}{R}, I = \frac{2B\ell v}{R}$

(2) $I_1 = I_2 = \frac{B\ell v}{3R}, I = \frac{2B\ell v}{3R}$

(3) $I_1 = I_2 = I = \frac{B\ell v}{R}$

(4) $I_1 = I_2 = \frac{B\ell v}{6R}, I = \frac{B\ell v}{3R}$

18. A coil of resistance R and inductance L is connected across an a.c. power supply of r.m.s. voltage V. The average power dissipated in the coil is,

(1) V^2/R (2) $\frac{V^2R}{(R^2+\omega^2L^2)}$

(3) $\frac{V^2}{\sqrt{(R^2+\omega^2L^2)}}$ (4) Zero

19. In an electromagnetic wave, the amplitude of electric field is 1 V/m. The frequency of wave is 5×10^{14} Hz. The wave is propagating along z-axis. The average energy density of electric field in Joule/m^3, will be

(1) 1.1×10^{11} (2) 2.2×10^{-12}
(3) 3×10^{-13} (4) 4.4×10^{-14}

20. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at relative speed of 15 m/s. The relative speed of the image of the second car as observed by the driver is

(1) $\frac{1}{15}$m/s (2) 10 m/s
(3) 15 m/s (4) $\frac{1}{10}$m/s

Integer Type Questions

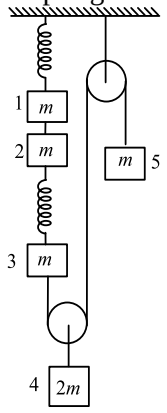
21. In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I_0 denotes the maximum intensity, then find the value of $\frac{3I_0}{I}$.

22. The wavelength of radiation emitted is λ_0, when an electron jumps from the third to the second orbit of hydrogen atom. For the electron jump from the fourth to second orbit of the hydrogen atom, the wavelength of radiation emitted will be λ' then the value of $\frac{27\lambda'}{10\lambda_0}$ is

23. Two waves given by equations $y_1 = 3\sin(100t - 20x)$mm and $y_2 = 4\cos(100t - 20x)$ mm interfere at a point then amplitude (in mm) of the resultant wave is

24. In a black body radiation certain temperature T_1, the wave length having maximum spectral intensity of radiation is 9000Å. when the temperature is increased from T_1 to T_2 the total radiation increases 16 times. The peak spectral radiation at T_2 is found to be capable of ejecting photoelectrons. The maximum kinetic energy of photoelectrons is the same as the energy of photon that one gets when one of electron in the M-shall of hydrogen atom jumps to L-shell. If the work function of the metal is ϕ then find the value of 15ϕ (in eV) (Use hc = 12400 eV-Å)

25. All the strings, springs and pulleys shown in figure are ideal. Initially the system is in equilibrium and blocks are at rest. Now, the upper spring is cut. The acceleration of block 1 just after cutting the upper spring will be (in m/s^2)



SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. Which among the following statement is incorrect?

(1) With excess ammonia, chlorine gives N_2 and NH_4Cl.

(2) With hot and conc. NaOH, Cl_2 gives a mixture of $NaClO_3$, NaCl and H_2O.

(3) HCl reacts with finely powdered iron to form ferric chloride.

(4) Electrification of Cl is more than that of F.

27. Which of the following compounds was unable to respond chromyl chloride Test?

(1) NaCl (2) $HgCl_2$

(3) $ZnCl_2$ (4) $CuCl_2$

28. In the following compounds of manganese what is the distribution of electrons on d-orbitals of manganese?

(i) $[Mn(H_2O)_6]^{2+}$ (ii) $[Mn(CN)_6]^{4-}$

(1) $t_{2g}^3 e_g^2$ in both

(2) $t_{2g}^5 e_g^0$ in both

(3) $t_{2g}^3 e_g^2$ in (i) and $t_{2g}^5 e_g^0$ in (ii)

(4) $t_{2g}^5 e_g^0$ in (i) and $t_{2g}^3 e_g^2$ (ii)

29. When acetone and chloroform are mixed, hydrogen bonding takes place between them. Such a liquid pair will cause.

(1) No deviations from Raoult's law

(2) Negative deviations from Raoult's law

(3) Positive deviations from Raoul'ts law

(4) None

30. For a reaction

$3A + B \rightarrow 2C$

$-\frac{d[A]}{dt} = K[A][B]$, the expression for $\frac{d[C]}{dt}$ will be:

(1) $\frac{2}{3}K[A][B]$ (2) $\frac{3}{2}K[A][B]$

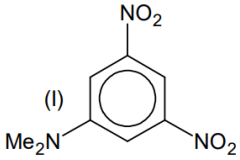
(3) $\frac{1}{3}K[A][B]$ (4) $\frac{1}{2}K[A][B]$

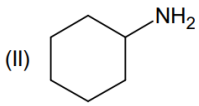
31. Equivalent conductance of saturated solution of $BaSO_4$ is 400 ohm^{-1} cm^2 eqv^{-1} and specific conductance is 8×10^{-5} ohm^{-1} cm^{-1}. Hence Ksp of $BaSO_4$ is

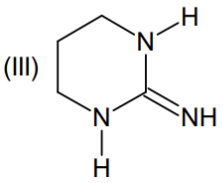
(1) 4×10^{-8} M^2 (2) 1×10^{-8} M^2

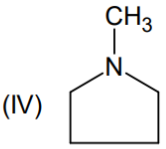
(3) 2×10^{-4} M^2 (4) 1×10^{-4} M^2

32. What is the correct order of basic strength of the following amines?









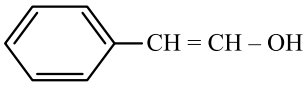
(1) (II) > (IV) > (III) > (I)

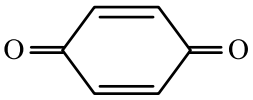
(2) (IV) > (I) > (III) > (II)

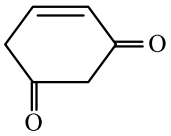
(3) (IV) > (I) > (II) > (III)

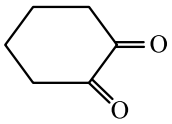
(4) (III) > (IV) > (II) > (I)

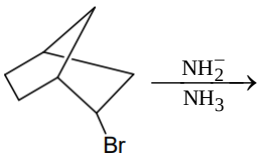
33. Tautomerism is not exhibited by

(1) 

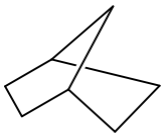
(2) 

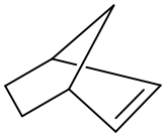
(3) 

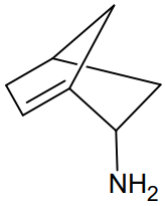
(4) 

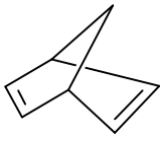
34. 

A (major product), A is

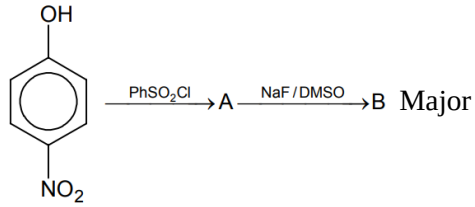
(1) 

(2) 

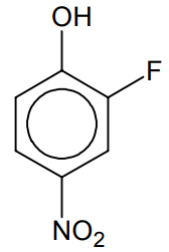
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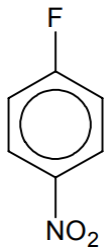
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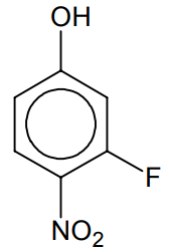
35. For the reaction sequence

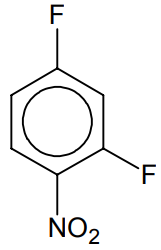


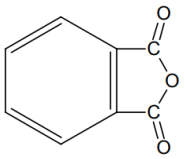
The compound B will be

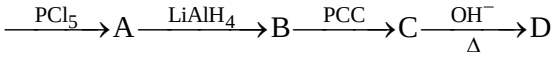
(1) 

(2) 

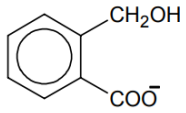
(3) 

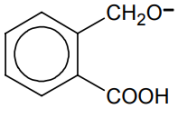
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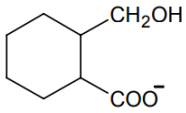
36. 

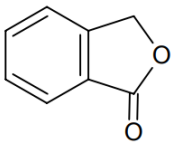


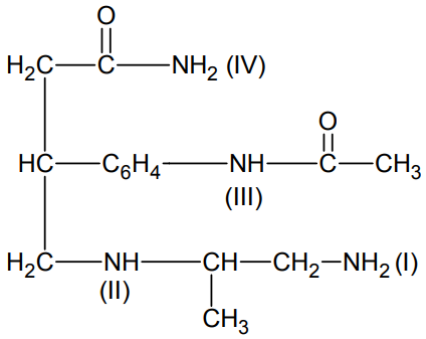
Product (D) is:

(1) 

(2) 

(3) 

(4) 

37. 

Arrange the above mentioned nitrogen atoms of the compound in decreasing order of basic nature?

(1) I > II > IV > III
(2) II > I > IV > III
(3) II > I > III > IV
(4) I > IV > III > II

38. The equilibrium constant of mutarotation of α-D-glucose to β-D-glucose is 1.8. If specific rotation of α-D-glucose is 112° and that of mixture is 59°; what is specific rotation of β-D-glucose?

(1) –112° (2) –59°
(3) +29.6° (4) –29.6°

39. The weight of a residue obtained by heating 2.76 g of silver carbonate is:
[Given atomic mass of Ag = 108]

(1) 2.76 g
(2) 2.98 g
(3) 2.16 g
(4) 2.44 g

40. Match the Column – I with Column – II:

List-I (Orbitals)		List-II (Quantum No.)	
A	2s	P	n = 2, ℓ = 0, m = 0
B	$2p_z$	Q	n = 2, ℓ = 1, m = 0
C	$4d_{x^2-y^2}$	R	n = 4, ℓ = 2, m = $\mp$2
D	$4d_{z^2}$	S	n = 4, ℓ = 2, m = 0

(1) A → P; B→ Q; C→R; D →S
(2) A → Q; B →P; C→R; D →S
(3) A → P; B → S; C→R; D→Q
(4) A → R; B → Q; C→P; D →S

41. **STATEMENT-1:** Bond energy of Cl – Cl bond is more than F – F bond.
and
STATEMENT-2: Shorter the bond length more is the bond energy.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(3) Statement-1 is True, Statement -2 is False.
(4) Statement-1 is False, Statement-2 is True.

42. The bond angle around central atom is highest in

(1) BBr_3
(2) CS_2
(3) SO_2
(4) SF_4

43. A metallic oxide contains 20% oxygen by mass. The equivalent weight of metal is:

(1) 40
(2) 32
(3) 16
(4) 48

44. Quick lime; (CaO) is produced by heating limestone $CaCO_3$ to drive off CO_2 gas:
$CaCO_3 \rightarrow CaO(s) + CO_2(g)$
$\Delta H = 180$ kJ mol^{-1}
$\Delta S = 150$ JK^{-1}
Assuming that variation of enthalpy change and entropy change with temperature to be negligible, which of the following is correct?

(1) Decomposition of $CaCO_3(s)$ is non-spontaneous at room temperature.
(2) Decomposition of $CaCO_3(s)$ becomes spontaneous when temperature is less than 927°C.
(3) Decomposition of $CaCO_3(s)$ becomes non-spontaneous when temperature is greater than 1200°C.
(4) All of the above.

45. Match the List-I with List-II:

List-I		List-II	
A	10 ml 0.1 M NaOH aq. solution is added to 10 ml 0.1 M HCl aqueous solution	P	Salt of weak acid and strong base
B	10 ml, 0.1 M NaOH aq. solution is added to 10 ml 0.1 M CH_3COOH aq. Solution	Q	Salt of weak base and strong acid
C	10 ml 0.1 M HCl aq. solution is added to 10 ml 0.1 M NH_3 aq. solution	R	pH of the solution will be 7
D	10 ml, 0.2 M HCl aq. solution to 10 ml, 0.1 M KOH aq. Solution	S	pH of the solution will be 1.33

(1) A → R; B → P; C → Q; D → S
(2) A → Q; B → P; C → S; D → R
(3) A → S; B → P; C → R; D → Q
(4) A → S; B → P; C → Q; D → R

Integer Type Questions

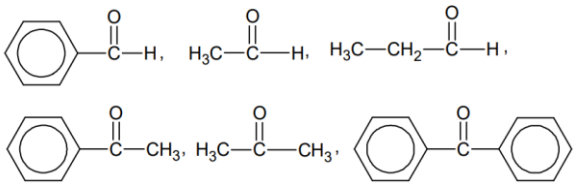
46. A complex of potassium, iron and cyanide ions $K_x[Fe(CN)_6]$ is 100% ionized at 1 molal. If its elevation in boiling points is 2.08°C, find out the value of 'x' in the complex.
(Given $K_b = 0.52$°C mol^{-1} kg)

47. Total number of cyclic isomers of molecular formula C_4H_7Cl is

48. The number of neutral oxides among the following oxides is
N_2O, NO, N_2O_3, NO_2, N_2O_5, CO, CO_2, SO_2 and CrO_3

49. The standard heat of combustion of $C_2H_2(g)$, C(graphite) and $H_2(g)$ are 310 Kcal mol^{-1}, 94 Kcal mol^{-1} and 98 Kcal mol^{-1} respectively. The heat of formation (in Kcal mol^{-1}) of $C_2H_2(g)$ is:

50. Among the following the number of compounds which react with Fehling solution is:



SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. The ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ have in common

(1) Centre only
(2) Centre, foci and directrices
(3) Centre, foci and vertices
(4) Centre and vertices only

52. If $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}.\vec{a}=0$ where $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}+3\hat{k}$, then $\vec{r}=$

(1) $\frac{1}{2}\left(\hat{i}+\hat{j}+\hat{k}\right)$ (2) $2\left(\hat{i}+\hat{j}+\hat{k}\right)$
(3) $2\left(-\hat{i}+\hat{j}+\hat{k}\right)$ (4) $\frac{1}{2}\left(\hat{i}-\hat{j}+\hat{k}\right)$

53. The coefficient of the term independent of x in the expansion of $\left[\frac{x+1}{x^{\frac{2}{3}}-x^{\frac{1}{3}}+1}-\frac{x-1}{x-x^{\frac{1}{2}}}\right]^{10}$ is

(1) 210
(2) 105
(3) 70
(4) 112

54. If $x = 2$ is a real root of cubic equation $f(x) = 3x^3 + bx^2 + bx + 3 = 0$ then the value of $\lim\limits_{x\to\frac{1}{2}}\frac{e^{f(x)}-1}{2x-1}=$

(1) $\frac{27}{4}$ (2) $\frac{-27}{4}$
(3) $\frac{-27}{8}$ (4) $\frac{27}{8}$

55. Let R be the relation on the set R of all real numbers defined by aRb if $|a - b| \leq 1$, then relation R is

(1) Reflexive and symmetric but not transitive
(2) Symmetric only
(3) Transitive only
(4) Symmetric and transitive but not reflexive

56. $f(x)$ is defined for $x > -1$, has a continuous derivate and satisfies $f(0) = 1$, $f'(0) = 0$; $(1 + f(x))\, f''(x) = 1 + x$. If x is positive then $f'(x)$ is

(1) Always positive
(2) Always negative
(3) Always non-negative
(4) None of these

57. Let $f(x) = (a - x^n)^{1/n}$ and $g(x) = \underbrace{fofofo\ldots f}_{2m \text{ times}}(x)$.

Then $\int \frac{dx}{(g(x))^n (1+(g(x))^n)^{1/n}}$ equals

(1) $\frac{n}{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}} + C$

(2) $\frac{1}{1-n}(1+x^{-n})^{\frac{n-1}{n}} + C$

(3) $\frac{1}{1+n}(1+x^n)^{\frac{n+1}{n}} + C$

(4) $\frac{-1}{n}(1-x^n)^{\frac{n}{n+1}} + C$

58. If $y = f(x)$ and $z = g(x)$, then $\frac{d^2y}{dz^2}$ equals

(1) $\frac{g'f'' - f'g''}{(g')^2}$ (2) $\frac{g'f'' - f'g''}{(g')^3}$

(3) $\frac{g'f' - f''g''}{(g')^3}$ (4) $\frac{g'f'' + f'g''}{(g')^3}$

59. The radius of any circle touching the lines $3x - 4y + 5 = 0$ and $6x - 8y - 9 = 0$ is (in units)

(1) 1.9 (2) 0.95
(3) 2.9 (4) 1.45

60. The differential equation of the system of circles touching the x-axis at origin is

(1) $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$

(2) $(x^2 - y^2)\frac{dy}{dx} + 2xy = 0$

(3) $(x^2 + y^2)\frac{dy}{dx} - 2xy = 0$

(4) $(x^2 + y^2)\frac{dy}{dx} + 2xy = 0$

61. If $\alpha, \beta, \gamma, \delta$ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k, then the value of $4\sin\frac{\alpha}{2} + 3\sin\frac{\beta}{2} + 2\sin\frac{\gamma}{2} + \sin\frac{\delta}{2}$ is equal to

(1) $2\sqrt{1-k}$ (2) $2\sqrt{1+k}$
(3) $2\sqrt{k}$ (4) None of these

62. If equation $x^3 - x + a^2 - a + \frac{2}{3\sqrt{3}} < 0$ is true for atleast one positive x, then a belongs to

(1) (0, 1)
(2) $(1, \infty)$
(3) $(2, 3\sqrt{3})$
(4) none of these

63. Number of solutions of the equation $\sqrt{3x-2} + \sqrt{4x+4} = \sqrt{5x+1} + \sqrt{4x-5}$ is

(1) 0 (2) 1
(3) 2 (4) 4

64. $\lim_{x\to\infty}\left[\sqrt{x^2+2x-1} - x\right]$ is equal to

(1) ∞
(2) $\frac{1}{2}$
(3) 4
(4) 1

65. If $0 \leq \arg z \leq \frac{\pi}{4}$, then the least value of $\sqrt{2}\,|2z - 4i|$ is

(1) 6 (2) 1
(3) 4 (4) 2

66. If α, β, γ be the roots of the equation $2x^3 + 3x^2 - 12x + 3 = 0$ and $A(\alpha, \beta, \gamma)$, $B(\beta, \gamma, \alpha)$, $C(\gamma, \alpha, \beta)$ represent vertices of a triangle ABC then the centroid of the triangle lies upon the line

(1) $x = y = z$
(2) $x = 2y = 3z$
(3) $x = -2y = 3z$
(4) $x = y = -2z$

67. If $a = \sqrt{\cot^{-1} x}, b = \cot^{-1} x$ and $c = (\cot^{-1} x)^2$ and $a > b > c$ then x lies in the interval

(1) (0, 1)
(2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(3) $(\cot 1, \infty)$
(4) $(-\cot 1, 1)$

68. If three one by one squares are selected at random from the chessboard, then the probability that they form the letter 'L' is

(Other orientations of ⌊ are also allowed e.g ⌋ or ⌈ or ⌉)

(1) $\frac{196}{{}^{64}C_3}$ (2) $\frac{49}{{}^{64}C_3}$

(3) $\frac{36}{{}^{64}C_3}$ (4) $\frac{98}{{}^{64}C_3}$

69. If $x^2+9y^2+25z^2=xyz\left(\frac{15}{x}+\frac{5}{y}+\frac{3}{z}\right)$, then

(1) x, y, z are in A.P

(2) x, y, z are in G.P

(3) $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ are in G.P

(4) $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ are in A.P

70. The number of ways of choosing triplet (x, y, z) such that $z > \max\{x, y\}$ and $x, y, z \in \{1, 2, \ldots, n, n+1\}$ can not be

(1) ${}^{n+1}C_3 + {}^{n+2}C_3$

(2) $\frac{1}{6}n(n+1)(2n+1)$

(3) $1^2+2^2+\ldots+n^2+(n+1)^2$

(4) $2({}^{n+2}C_3) - {}^{n+1}C_2$

Integer Type Questions

71. If $f(x)+f\left(1-\frac{1}{x}\right)=1+x\forall x\in R-\{0,1\}$.

The value of $4f(2)$ is equal to _____.

72. Given $A = \cos^2\theta(\cos^2\theta + 1) + 2\sin^2\theta$. Then for all θ, $[A]$ can assume values whose sum is _____. (Where [.] denotes the greatest integer function).

73. If $3n$ be the sum of all possible integer values of 'p' for which $\left|x+\frac{1}{x}-3\right| = p-3$ has exactly two distinct solutions then the value of n is _____.

74. For positive t, let α, β be the roots of the quadratic equation $x^2 + t^2x - 2t = 0$. If minimum value of $\int_{-1}^{2}\left(\left(x+\frac{1}{\alpha^2}\right)\left(x+\frac{1}{\beta^2}\right)+\frac{1}{\alpha\beta}\right)dx$ is $\sqrt{p}+q$ where p is a rational number and $q \in N$, then the value of $[p] + q$ is (where [.] denotes G.I.F)

75. A species has an initial population 4^{10}. At the end of the 1st day, the population increases by 50 %. At the end of the 2nd day it decreases by the same percentage. If the process continues in the same pattern, and let $5x$ be the number of days required for population to be 3^{10}, then x is _____.



PW Web/App - https://smart.link/7wwosivoicgd4

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